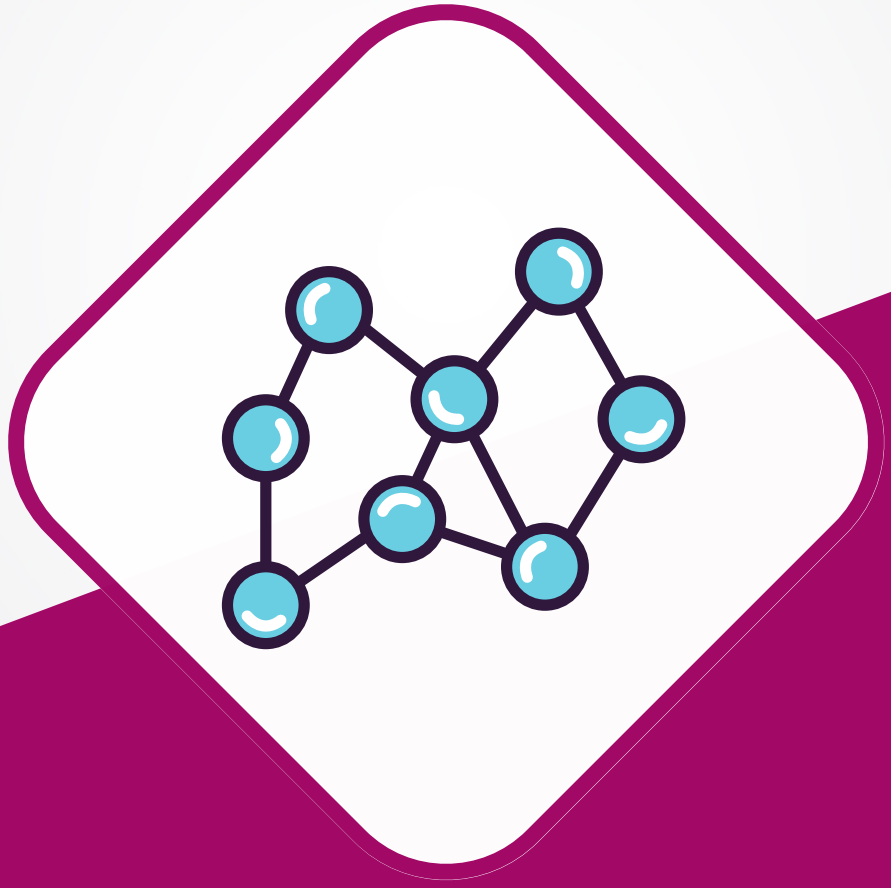
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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JEE (MAIN+ADVANCED)

Inorganic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Chemical Bonding

English Medium

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CHEMICAL BONDING

FLOWCHART TO CHEMICAL BONDING

Why a chemical bond is formed ?

- Lewis-Kossel approach (Octet)
  - Lewis Structure
  - Formal Charge
  - Resonance
- Decrease in Potential energy
  - Ionic attraction
  - Attraction between Nucleus & electron cloud

Type of Bond

Forces of attraction
(1) Hydrogen Bonding
(a) Inter molecular H-bonding
(b) Intra molecular H-bonding
(2) Vander Waals Force
(3) Ion-dipole
(4) Ion induced dipole

Typical Bonds
(1) Odd Electron bond
(2) Back Bonding
(3) Multicentre bond
(a). Banana bond ($3c–2e^-$)
(b). ($3c–4e^-$) bond

Chemical bond

Ionic Bond
(1) Electrovalency: or Loss & Gain of electrons
(2) Favourable Conditions
(a) Low I.E.
(b) High E.A.
(c) High Lattice Energy
(3) General Characteristics
(a) Good electrolytes
(b) Hard, Crystalline , Brittle
(c) High M.P./B.P.
(d) Fast reactions in polar solvent
(4) Inert Pair Effect
(5) Covalent character in ionic compounds: Fajan's rule

Covalent Bond
(1) Sharing of electrons or Covalency
(2) Favourable Conditions
3, 4, 5, 6 ,7 valence electrons at central atom
(3) General Characteristics
(a) Low M.P. & B.P.
(b) Generally poor/ bad conductor of electricity except graphite
(4) Theories
(a) Electronic theory of valency
(b) VBT(hybridisation)
(c) MOT
(5) Due to E.N. difference 'Dipole Moment'

Co-ordinate Bond
(1) Lewis acid
(2) Lewis base

Metallic Bond

Note : Hydrogen bond is an extreme manifestation of dipole dipole interaction.

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VANDER WAAL'S FORCES

⇒ These are the weakest type of inter molecular forces that exist among the chemical species which being significant change in physical properties.

⇒ These are non-directional, non-valence cohesive forces. These attractive forces being played between the two molecules are independent of the presence of other molecules.

⇒ Solid, liquid or gaseous states of many molecules are explained on the basis of inter molecular forces other than covalent, ionic or metallic bonds. Although inert gases do not form any type of bond but may exist in liquid and solid states. This shows that the atoms of inert gases are attracted by each other through some type of inter molecular forces. These intermolecular forces are called Vander Waals forces.

Types of Vander Waal's Forces

(1) ***Dipole-dipole interaction (Keesom forces) :*** The force of attraction between the oppositely charged poles of two polar molecules (for example : H_2S, HCl, PH_3 etc.) is called dipole-dipole attraction.

(2) ***Dipole-induced dipole interaction (Debye forces) :*** This type of cohesive forces occurs in a mixture of polar and non polar molecules. The former induce polarity in non polar molecules by disturbing their electron system. for example force of attraction between Cl_2 and H_2O.

(3) ***Instantaneous dipole-Induced dipole interaction (Dispersion forces/London forces) :*** The weak intermolecular forces operating in similar non polar gaseous molecules are called London forces. These forces are very weak in nature and exists only at low temperature. For example weak intermolecular forces in F_2, Cl_2, N_2, molecules and in noble gasses.

(Note :- London forces present in both polar and non polar species)

Other Weak Interactions

(1) ***Ion-dipole interaction :*** Polar molecules are attracted by ions. The negative pole is attracted by cation and positive pole attracted by the anion. This type of attraction is called ion dipole attraction, ion-dipole attraction is observed generally in the process of solvation when sodium chloride (Na^+ Cl^-) is dissolved in water because negative poles of water aggregate around Na^+ ions and positive poles around Cl^- ions.

(2) ***Ion-induced dipole interaction :*** When non polar molecules come in contact with ions, its electron cloud gets polarised and the oppositely charged end of it is attracted by the ion. For example attraction between I^- and I_2 molecule.

HYBRIDISATION

The phenomenon of intermixing of the orbitals of slightly different energies so as to redistribute their energies, resulting in the formation of new set of orbitals of equivalent energies and shape. For example when one 2s and three 2p-orbitals of carbon hybridise, there is the formation of four new sp^3 hybrid orbitals.

Salient features of hybridisation: The main features of hybridisation are as under :

1. The number of hybrid orbitals is equal to the number of the atomic orbitals that get hybridised.

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2. The hybridised orbitals are equivalent in energy and shape.
3. The hybrid orbitals are more effective in forming stable bonds than the pure atomic orbitals.
4. These hybrid orbitals are directed in space in some preferred direction to have minimum repulsion between electron pairs and thus a stable arrangement. Therefore, the type of hybridisation indicates the geometry of the molecules.

Important conditions for hybridisation

(i) The orbitals present in the valence shell of the atom are hybridised.

(ii) The orbitals undergoing hybridisation should have almost equal energy.

(iii) Promotion of electron is not essential condition prior to hybridisation.

(iv) It is not necessary that only half filled orbitals participate in hybridisation. In some cases, even filled orbitals of valence shell take part in hybridisation.

Types of Hybridisation

There are various types of hybridisation involving s, p and d orbitals. The different types of hybridisation are as under :

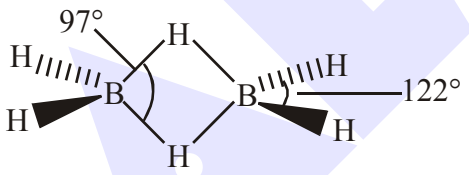
(i) *sp hybridisation* (ii) *sp^2 hybridisation* (iii) *sp^3 hybridisation*

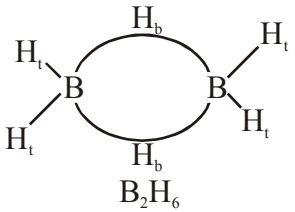
(iv) *sp^3d hybridisation:* (v) *sp^3d^2 hybridisation:* (vi) *sp^3d^3 hybridisation:*

BRIDGE BONDING

There are many compounds in which some electron deficient bonds are present apart from normal covalent bonds or coordinate bonds which are 2c-2e$^-$ bonds(two centre two electron bonds). These electron deficient bonds have less number of electrons than the expected such as three centre-two electron bonds (3c-2e) which are present in diborane B_2H_6, $Al_2(CH_3)_6$, BeH_2(s) etc.

bonding in B_2H_6(diboarane)



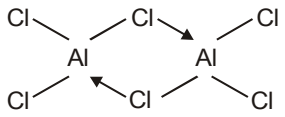


B_2H_6

The structure of diborane containing four terminal(t) and two bridging(b) hydrogen atoms. The model determined by molecular orbital theory indicates that the bonds between boron and the terminal hydrogen atoms are conventional 2c-2e$^-$ covalent bonds. The bonding between the boron atoms and the bridging hydrogen atoms is, however different from that in molecules such as hydrocarbons. Having used two electrons in bonding to the terminal hydrogen atoms, each boron has one valence electron remaining for additional bonding. The bridging hydrogen atoms provide one electron each. Thus the B_2H_2 ring is held together by four electrons, an example of 3c-2e$^-$bonding. This type of bond is sometimes called as **'banana bond'**. Group 13, gallium is known to form a similar compound, digallane, Ga_2H_6.

But Al_2Cl_6 have covalent bond only and there is no electron deficient bonding as depicted in the given structure.

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(Al – Cl – Al is 3c – 4e bond)

INERT PAIR EFFECT

In p-block elements the stability of the lower oxidation state increases on descending the group. Because increased effective nuclear charge holds ns electrons tightly due to poor shielding effect of inner d & f orbitals and thereby, restrict their (ns electrons) participation in bonding only np electrons take part in bond formation. As a result of this, +1 oxidation state of Tl is more stable than it's +3 oxidation state. Pb shows +2 stable oxidation state and Bi shows +3 stable oxidation state.

For example :

Group 13	**Group 14**
B (+3)	C (+4)
Al (+3)	Si (+4)
Ga (+3), (+1)	Ge (+4), (+2)
In (+3), (+1)	Sn (+4), (+2)
Tl (+3), (+1)	Pb (+4), (+2)

Order of stability : $Tl^{+1} > In^{+1} > Ga^{+1}$ (due to inert pair effect)

Order of stability : $Pb^{+2} > Sn^{+2} > Ge^{+2}$ (due to inert pair effect)

Ex. $PbCl_4$ is stable at room temperature whereas PbI_4 doesn't exist.

Sol. Due to inert pair effect Pb(+4) is less stable than Pb(+2). Hence it is very good oxidant.

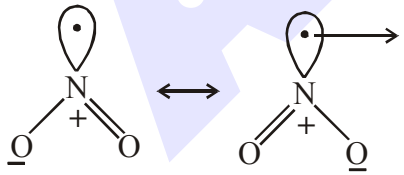
$Pb(IV) + 2e^- \longrightarrow Pb(II)$

Reducing abilities of halides follows the sequence

$I^- > Br^- > Cl^-$

STRUCTURE OF ODD ELECTRONIC SPECIES

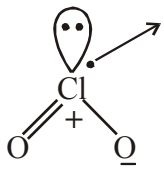
(1) NO_2



Free electron occupies the one sp^2 hybrid orbital.

(2) ClO_2

Structure :-

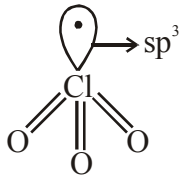


The free electron resides at the 3d-orbital of Cl-atom.

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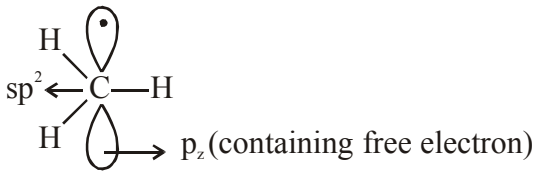
(3) ClO_3 :

Structure :



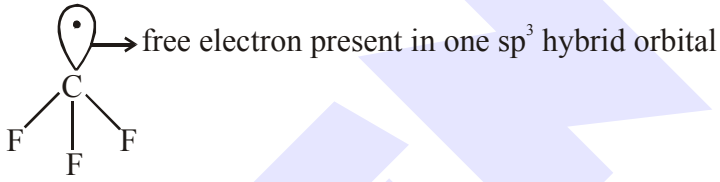
(4) $\dot{C}H_3$:

Structure :



(5) $\dot{C}F_3$:

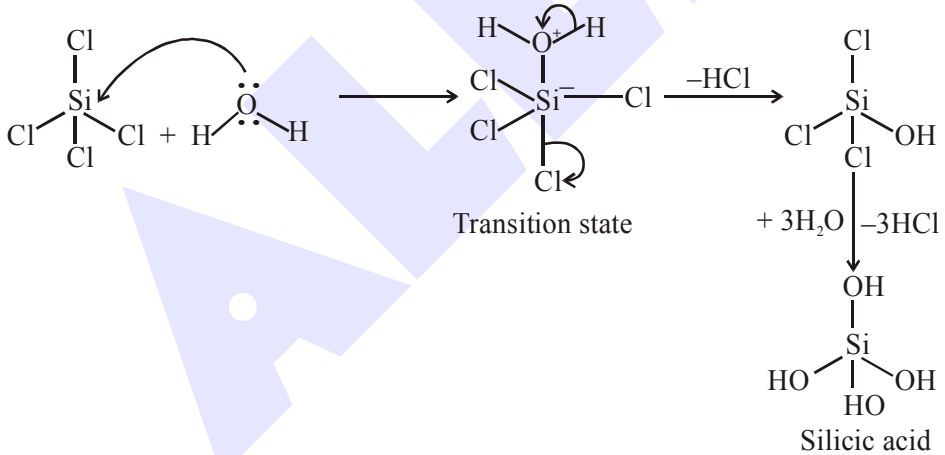
Structure :



HYDROLYSIS :

In hydrolysis of covalent molecules the nucleophilic centre of molecule is replaced by OH^- group of water generally through nucleophilic substitution reaction.

Ex. Hydrolysis of $SiCl_4$



Note : CCl_4, NF_3, is inert towards hydrolysis due to the absence of d orbital, but under drastic condition these molecules under goes hydrolysis.

$$CCl_4 + H_2O \xrightarrow{\text{superheated}} COCl_2 + 2HCl$$

Note : Hydrolysis of XeF_2 & XeF_4 takes place through with redox reaction.

$$XeF_2 + H_2O \longrightarrow Xe + 2HF + \frac{1}{2}O_2$$

$$3XeF_4 + 6H_2O \longrightarrow 2Xe + XeO_3 + 12HF + \frac{3}{2}O_2$$

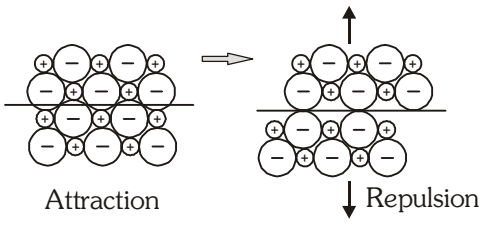
$$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$$

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IONIC COMPOUNDS

Properties of ionic compound

(a) Physical state : Ionic compounds are hard, crystalline and brittle due to strong electrostatic force of attraction. Brittleness → {Same charged ions comes nearer. So they repell each other}



(b) Isomorphism : The phenomenon of different ionic compounds, having same crystal arrangement of ions is termed as isomorphism

Condition of Isomorphism :

(i) Same charge on cation & anion between isomorphs

(ii) Same radius ratio range of cation & anion between isomorphs

(iii) Same number of water of crystalization between isomorphs

Ex. (i) $ZnSO_4 \cdot 7H_2O$, $FeSO_4 \cdot 7H_2O$ are isomorphous

(ii) All alums are isomorphous

(c) Boiling point and melting point : Ionic compounds have high boiling point and melting point due to strong electrostatic force of attraction among oppositely charged ions.

(d) Conductivity : It depends on ionic mobility.

(i) In solid state - No free ions - Bad conductor of electricity.

(ii) In fused state or aqueous solution Due to free ions - Good conductor of electricity.

Conductivity order : Solid state < fused state < Aqueous solution

(e) Solubility :

Ionic compounds are more soluble in polar solvalents and less soluble in non polar solvents.

• Solubility of ionic compounds in water mainly depends upon hydration energy & lattice energy.

Q. Why does the solubility of alkaline earth metal hydroxides in water increase down the group?

Ans. Among alkaline earth metal hydroxides, the anion being common the cationic radius will influence the lattice enthalpy. Since lattice enthalpy decreases much more than the hydration enthalpy with increasing ionic size, the solubility increases as we go down the group.

Q. Why does the solubility of alkaline earth metal carbonates and sulphates in water decrease down the group?

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Ans. The size of anions being much larger compared to cations, the lattice enthalpy will remain almost constant within a particular group. Since the hydration enthalpies decrease down the group, solubility will decrease as found for alkaline earth metal carbonates and sulphates.

FAJAN'S RULE

Just as all the covalent bonds have some partial ionic character, the ionic bonds also have partial covalent character. The partial covalent character of ionic bonds was discussed by Fajans in terms of the following rules :

- The smaller the size of the cation and the larger the size of the anion, the greater the covalent character of an ionic bond.
- The greater the charge on the cation, the greater the covalent character of the ionic bond.
- For cations of the same size and charge, the one, with electronic configuration $(n-1)d^n ns^0$, typical of transition metals, is more polarising than the one with a noble gas configuration, $ns^2 np^6$, typical of alkali and alkaline earth metal cations.
- The cation polarises the anion, pulling the electronic charge toward itself and thereby increasing the electronic charge between the two. This is precisely what happens in a covalent bond, i.e., buildup of electron charge density between the nuclei. The polarising power of the cation, the polarisability of the anion and the extent of distortion (polarisation) of anion are the factors, which determine the per cent covalent character of the ionic bond.

$\Rightarrow$ Polarisation power of a cation is usually called ionic potential or charge density.

$$\text{Ionic potential } \phi \text{ (phi)} = \frac{\text{Charge on cation}}{\text{Size of cation}}$$

APPLICATION OF THE CONCEPT OF POLARISATION :

(a) To compare the covalent and ionic character of molecule
(b) To compare the nature of oxide
(c) To compare the electrical conductivity of ionic comopounds
(d) Tendency of the formation of complex compounds
(e) To compare the thermal stability of metal salts
(f) To compare the intensity of colour of compounds
(g) To compare the solubility of heavier metal halide in water.

MOLECULAR ORBITAL THEORY (MOT)

Given by Hunds & Mulliken

(a) Two atomic orbital come nearer & then overlap each other to form two molecular orbitals (MO)
(b) Combination of atomic orbital (AO) forms molecular orbital (MO)

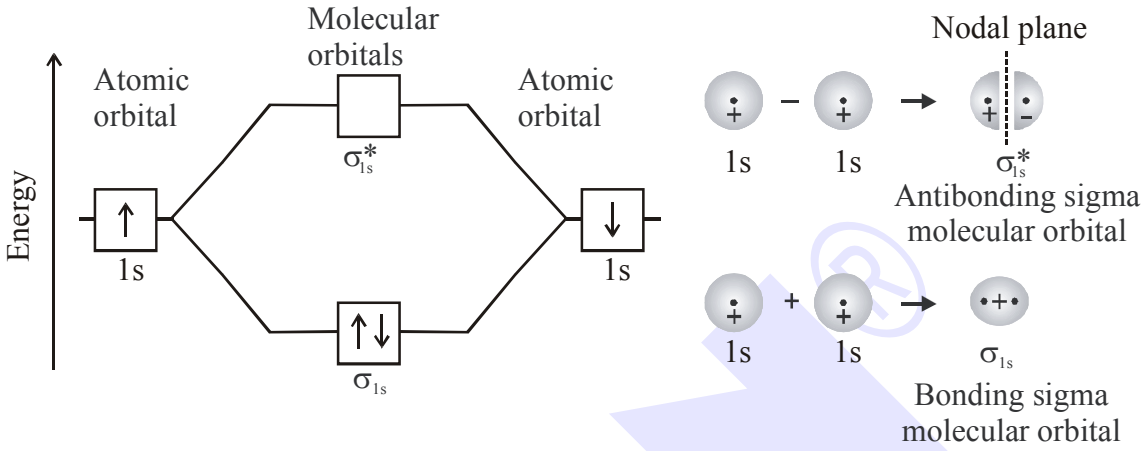
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Types of molecular orbitals

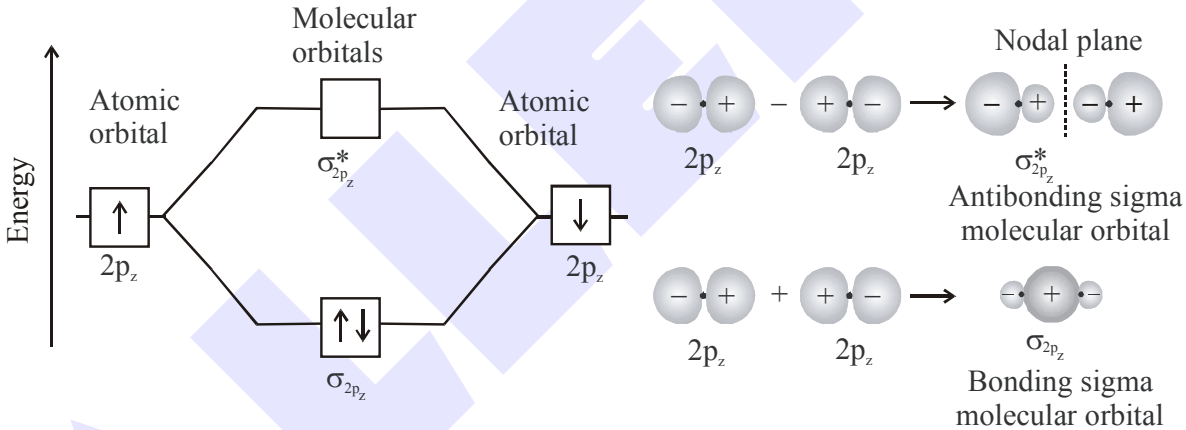
Molecular orbitals of diatomic molecules are designated as σ(sigma), π(pie), δ(delta) etc.

In this nomenclature, the **sigma (σ) molecular orbitals are symmetrical around the inter molecular axis (assumed to be z-axis) while pi (π) molecular orbitals are not symmetrical.**

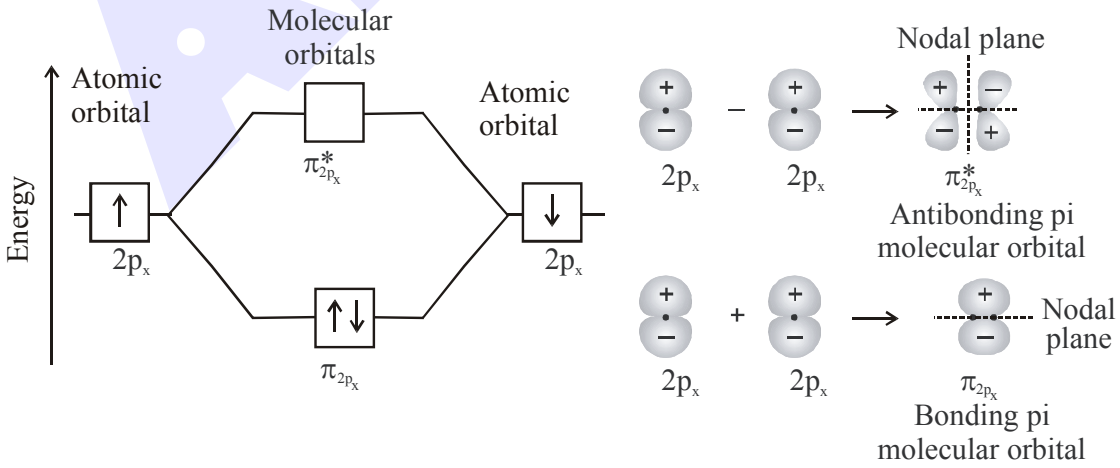
(1) s-s combination of orbitals



(2) p-p combination of orbitals(end to end overlap)

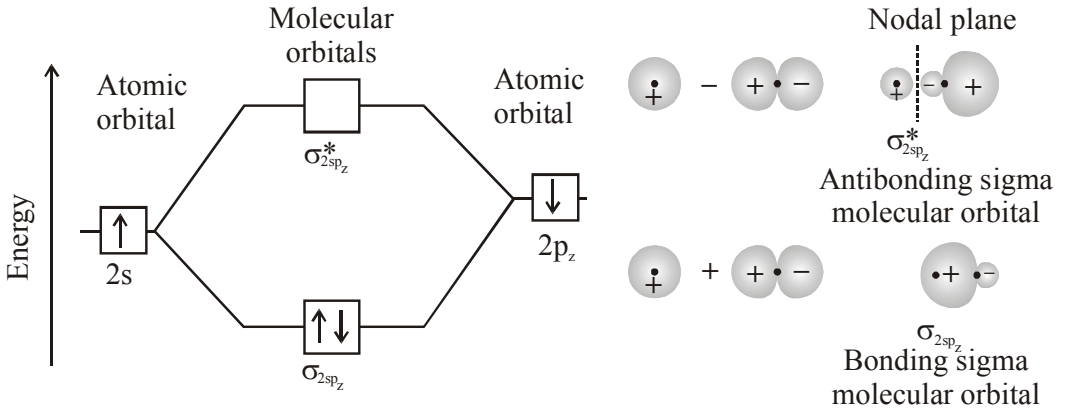


(3) p-p combination of orbitals (side by side overlap)



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(4) s-p combination of orbitals



(c) Energy of BMO < Energy of ABMO.

(d) Molecular orbitals can be filled by electrons according to Aufbau, Hund's, Pauli's principle.

(e) Energy order of the molecular orbitals of homonuclear di-atomic molecules.

Note : Molecular orbital energy diagram for up to N_2 (molecule having ≤ 14 electrons)

$$\sigma_{1s} < \sigma^*_{1s} < \sigma_{2s} < \sigma^*_{2s} < \pi_{2p_x} = \pi_{2p_y} < \sigma_{2p_z} < \pi^*_{2p_x} = \pi^*_{2p_y} < \sigma^*_{2p_z}$$

Note : Molecular orbital energy diagram for O_2 and F_2 (molecule having > 14 electrons)

$$\sigma_{1s} < \sigma^*_{1s} ; < \sigma_{2s} < \sigma^*_{2s} < \sigma_{2p_z} < \pi_{2p_x} = \pi_{2p_y} < \pi^*_{2p_x} = \pi^*_{2p_y} < \sigma^*_{2p_z}$$

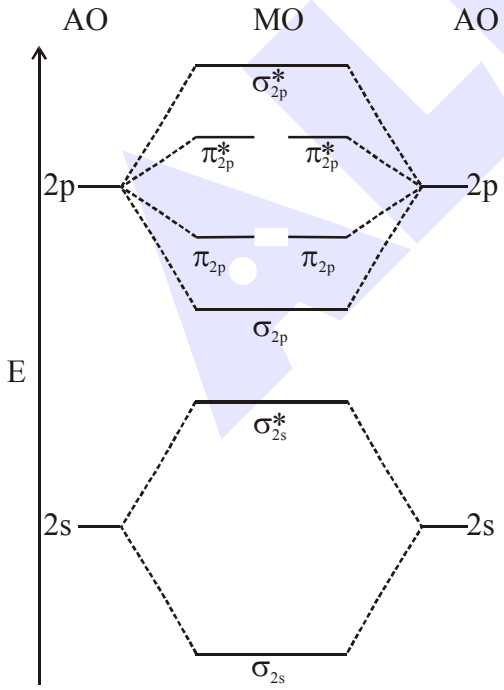
$\sigma *$, $\pi *$ = antibonding molecular orbital

σ, π = bonding molecular orbital

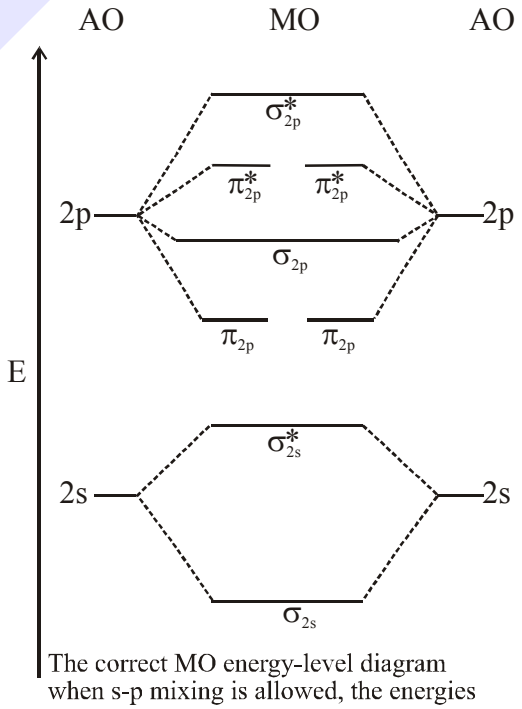
Ex. Why molecular orbitals have different order of energy in N_2 & O_2 ?

Sol. s-p mixing

Hint



The correct MO energy-level diagram when s-p mixing is not allowed.



The correct MO energy-level diagram when s-p mixing is allowed, the energies of the σ_{2p} and π_{2p} orbitals are reversed.

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Bond Order

Bond order can be defined as :

$$\textbf{Bond order} = \frac{N_b - N_a}{2}$$

N_b = No. of electron in bonding MO's

N_a = No. of electron in antibonding MO's

- If bond order = 0, it means species does not exist.
- Bond order of 1, 2 & 3 corresponds to a single bond, double & triple bond respectively.
- Bond order ↑ stability of molecule ↑ bond length ↓

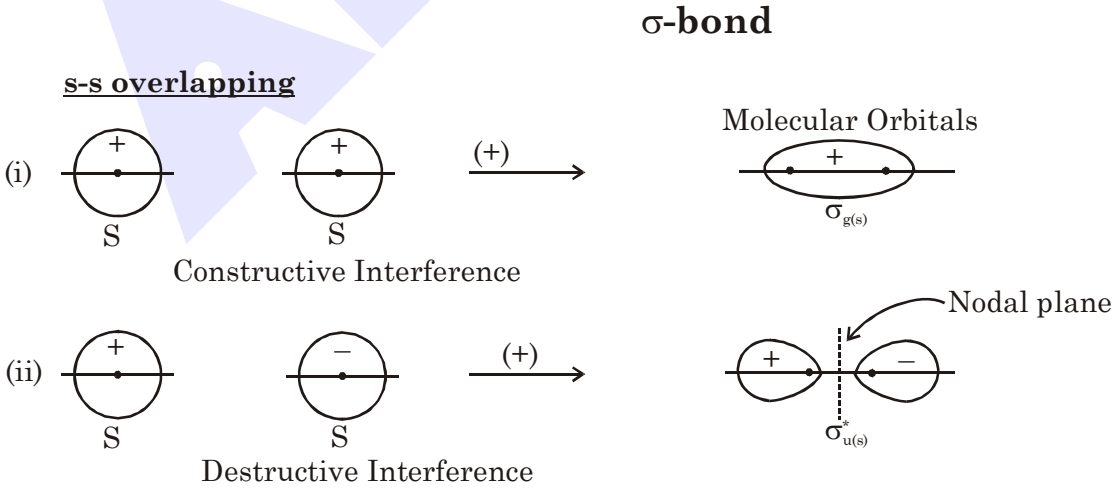
Magnetic behaviour

- If the molecule has one or more unpaired electron, it will be paramagnetic,
- If all the electrons are paired it will be diamagnetic.
- Magnetic strength can be calculated by using spin only formula of magnetic moment (μ).
- $\mu = \sqrt{n(n+2)}$ B.M. (where n = number of unpaired electron)

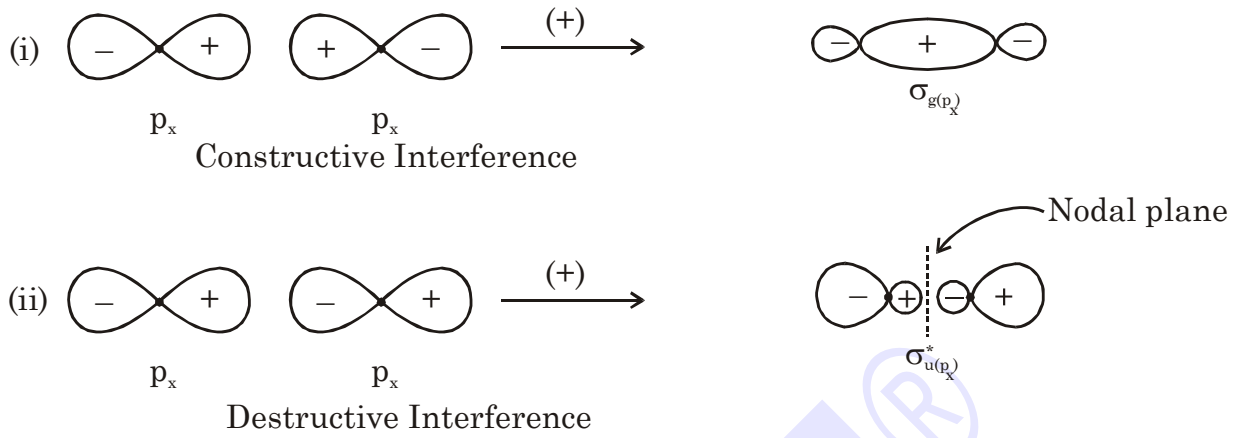
Ex. H_2 = Configuration : $\sigma^2_{(1s)}$, $\sigma^{*0}_{(1s)}$

Bond order = $\frac{N_b - N_a}{2} = \frac{2-0}{2} = 1$, Hence H – H (dimagnetic)

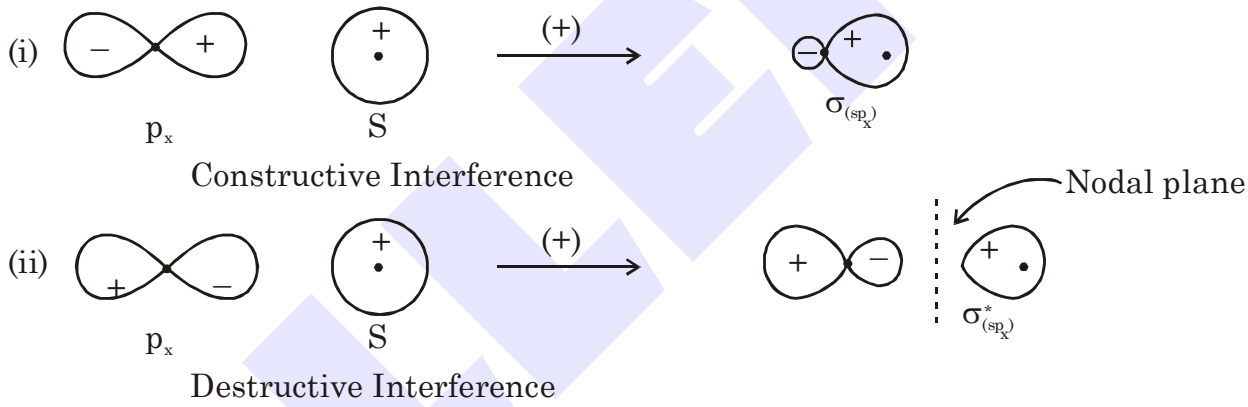
LINEAR COMBINATION OF ATOMIC ORBITALS (L.C.A.O.)



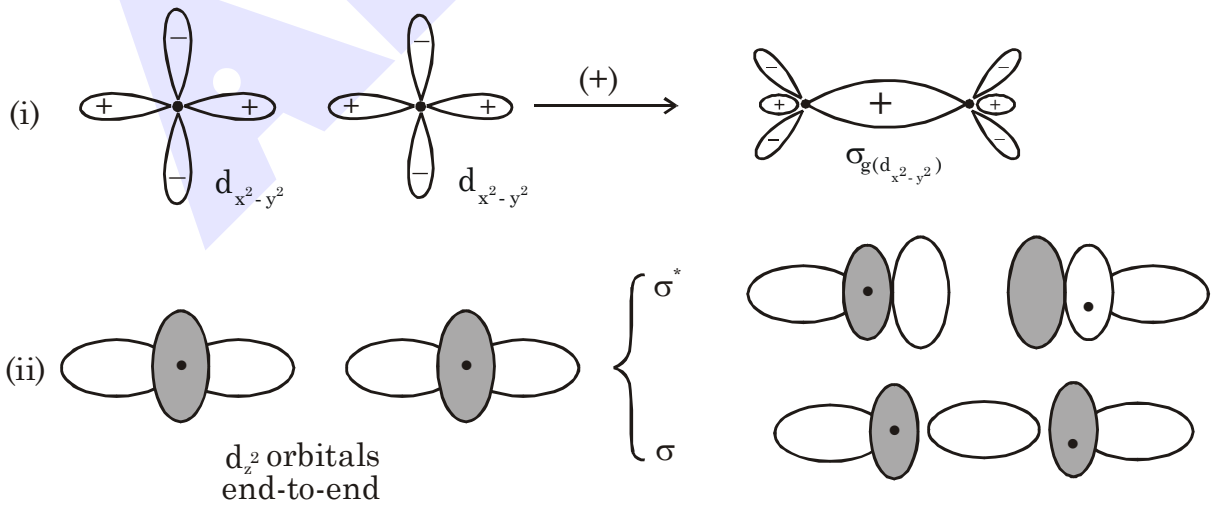
p-p overlapping



s-p overlapping



d-d overlapping

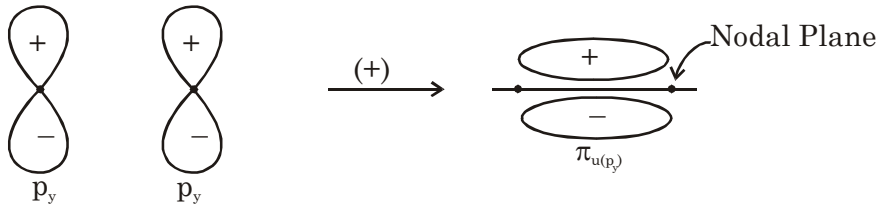


node06\B0BC-BD\Kota\JEE(Advanced)\Enfuse\Chem\Module\Mod.#Chemical Bonding-Co-chem & Metallurgy\(OC)\Eng\01-CB\01-Theory\p65

π-bond

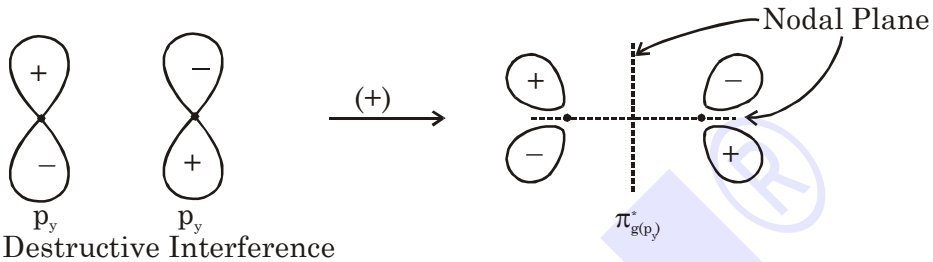
p-p overlapping

(i)



Constructive Interference

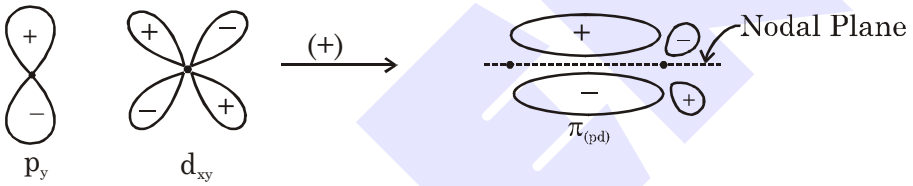
(ii)



Destructive Interference

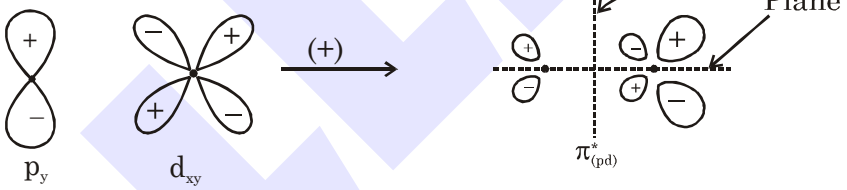
p-d overlapping

(i)



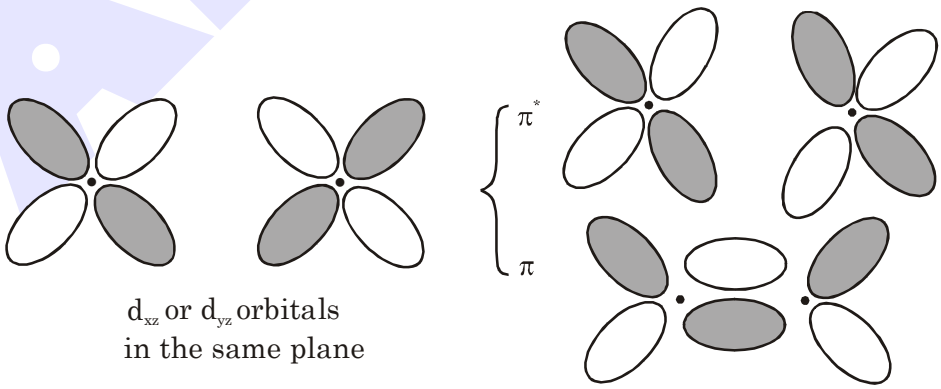
Constructive Interference

(ii)



Destructive Interference

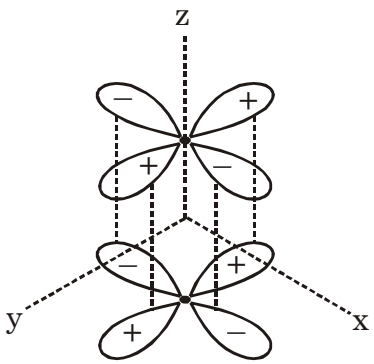
d-d overlapping

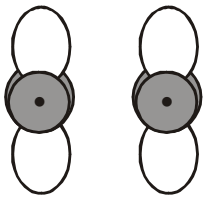


d_{xz} or d_{yz} orbitals
in the same plane

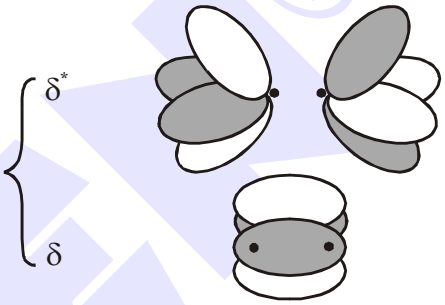
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #Chemical Bonding-Co-chem & Metallurgy\(OC)\Eng\01-CB\01-Theory.p65

δ-bond

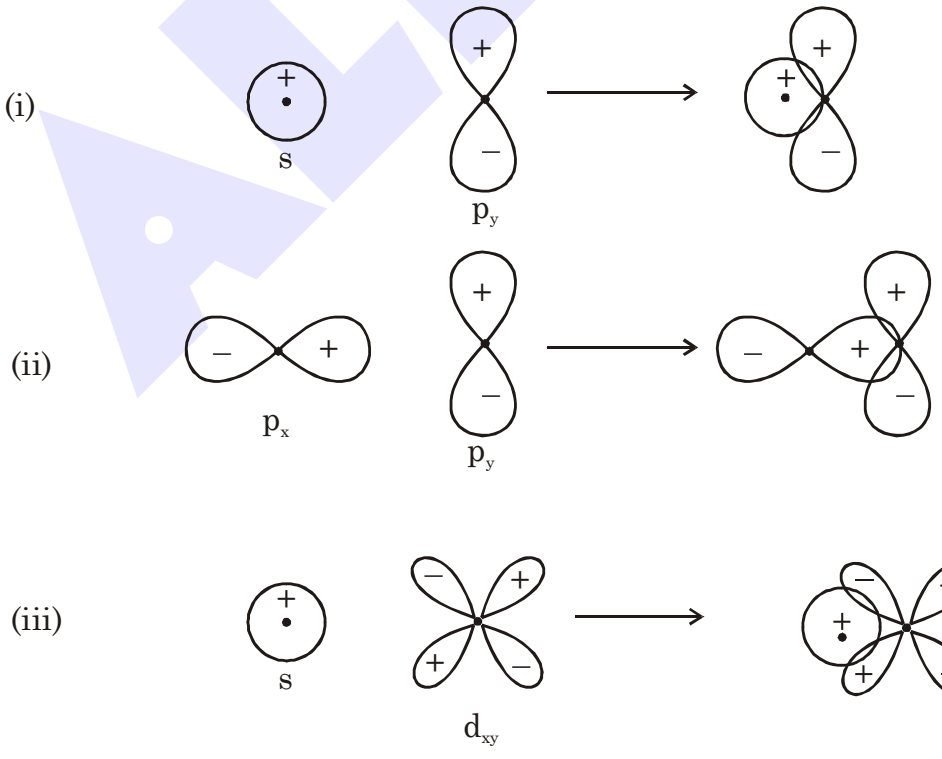




$d_{x^2-y^2}$ or d_{xz} orbitals
in parallel planes



non bonding molecular orbitals



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